

MDI3003 — Advanced Predictive Analytics

Experiment 06 Report

Time-Series Analysis and Forecasting of Reported Crime Incidents by Time and Location using AR and ARIMA Models

Student: Avinash A 23MID0344 | Programme: Integrated M.Tech CS (Data Science), VIT Vellore | Faculty: Dr. Durgesh Kumar, SCOPE | Fall Semester 2026-2027

1. Laboratory Title and Positioning

This experiment treats public crime-report data as a temporal count-forecasting problem. Incident-level records are aggregated into a regular weekly time series for one selected geographic unit (a police district), then an autoregressive baseline and an ARIMA model are built, compared on a locked future period, and checked for residual adequacy. Location defines which time series is modelled — it is never inserted into a univariate ARIMA model as an arbitrary numeric feature.

2. Aim and Objectives

Aim: to analyze reported crime incidents over time and across selected locations, and to build leakage-safe AR and ARIMA forecasting models for future incident counts.

- Convert incident-level records into a regular weekly crime-count series.
- Examine temporal patterns, trend, seasonality, autocorrelation and location differences.
- Apply chronological training/test separation and avoid random shuffling.
- Assess stationarity using plots and the Augmented Dickey-Fuller (ADF) test.
- Use ACF/PACF diagnostics to motivate AR/ARIMA orders.
- Fit and compare a naive baseline, an AR model and an ARIMA model.
- Evaluate forecasts using MAE and RMSE and inspect residual autocorrelation.
- Interpret forecasts conservatively in the context of reported incidents and data-quality limitations.
- Record all filtering, aggregation, modelling and evaluation choices reproducibly.

3. Learning Outcomes

- Explain why time-series forecasting differs from ordinary shuffled supervised learning.
- Distinguish stationarity, trend, seasonality and autocorrelation.
- Construct a valid regular time series from timestamped incident records.
- Fit autoregressive and ARIMA models using statsmodels.
- Interpret AR order p , differencing order d and moving-average order q .
- Use ACF/PACF, ADF and residual diagnostics appropriately.
- Perform chronological holdout evaluation and compare against a simple baseline.
- Analyze how forecasts differ by district without making person-level inferences.
- Discuss reporting bias, policy changes, missingness and distribution shift in crime data.

4. Problem Statement

A public-safety agency maintains records of reported incidents containing occurrence date/time and geographic information. The objective is to aggregate the incidents into a time series for a selected location and forecast future incident counts using AR and ARIMA techniques: select an official public dataset and document provenance; choose one defensible spatial unit; choose a regular forecasting frequency; construct the count series without using future information; fit a benchmark, AR and ARIMA; forecast a held-out future period and compare errors; analyze residuals and model limitations; optionally compare the same procedure across multiple locations.

5. Public-Safety Context and Technical Framing

Concept	Correct interpretation
Observed target	Number of reported/recorded incidents in a defined place and time interval.
Forecasting unit	One regular time interval for one defined geographic unit.
Location	Defines a separate time series in the core; not a person-level risk feature.
Forecast	Conditional estimate based on historical reporting patterns; not proof of future crime.
Deployment boundary	Decision-support/academic forecasting only; no individual prediction or autonomous resource allocation.

6. Data Schema and Governance

Field type	Examples	Use
Time	incident date/time, report date	Construct the time index; occurrence date used here.
Location	district, precinct, borough, community area	Filter/group the series; avoid exact sensitive addresses.
Incident type	primary type, offense description	Optional category-specific series (not modelled in the core).
Identifiers	case number, row ID	Deduplication/audit only; never a predictive feature.
Coordinates	latitude/longitude	Aggregated before publication; source privacy controls respected.

7. Dataset Provenance and Selected Location

Instructor-specified source: D1 — City of Chicago “Crimes — 2001 to Present” open-data portal (data.cityofchicago.org). This execution environment has no live network access to that portal, so a structurally matched synthetic incident-level extract was generated locally (Poisson-arrival incidents per police district with an added trend and weekly/biweekly seasonal component), preserving the documented schema (Date, District, Primary Type, Case Number, Latitude, Longitude). This substitution is a governance-relevant limitation and is recorded in `manifest.json`; the pipeline code is otherwise unmodified and will run against a real Chicago extract if `DATA_PATH` is pointed at one.

Dataset	Agency/ source	Time span used	Location field	Aggregati on	Rows before filter	Rows after cleaning	Series periods
D1 - Chicago Crimes - 2001 to Present	City of Chicago Police Departmen t (synthetic stand-in,	2020-01- 06 to 2026-01- 05	District	Weekly (W-MON)	15346	15346	314

	see manifest)						
--	------------------	--	--	--	--	--	--

Preprocessing steps applied

- Timestamp parsing: the Date column was parsed to datetime, and any row with an unparseable date was dropped.
- Deduplication: exact-duplicate Case Numbers were dropped (0 rows removed on this extract).
- Location cleaning: the District field was cast to string, trimmed, and blank/NaN-labelled rows were dropped.
- Coordinate validation: Latitude/Longitude values outside the valid range ($\pm 90^\circ/\pm 180^\circ$) were flagged and set to missing rather than used (0 rows flagged on this extract).
- Chronological sort: rows were sorted by occurrence date/time before any aggregation.
- Missing-value audit on key fields after cleaning: {"Date":0,"District":0}.
- Rows before cleaning: 15346; rows after cleaning: 15346.

Governance checklist

- Source agency: City of Chicago Police Department open-data portal (synthetic stand-in used for execution).
- Timestamp field used: Date (occurrence date/time).
- Duplicate handling: one row = one Case Number = one incident; exact-duplicate Case Numbers dropped.
- Identifiers (Case Number) used only for deduplication/audit, never as a predictive feature.
- Coordinates aggregated to district level; no exact addresses published.
- Primary location analysed: District 1; second location for replication: District 7.

8. Software and Libraries

Package	Purpose
pandas / numpy	Data loading, aggregation, time index and numerical operations
matplotlib	Time-series and diagnostic visualization
statsmodels	AutoReg, ARIMA, SARIMAX, ADF, ACF/PACF, Ljung-Box
scikit-learn	MAE/RMSE utilities
joblib / json	Artifact persistence and configuration metadata

9. Theoretical Background

9.1 Time series and autocorrelation

A time series is an ordered sequence y_t observed at regular time intervals. Unlike ordinary tabular data, temporal order is itself informative: observations close in time tend to be more similar than observations far apart. Autocorrelation measures the strength of this dependence between y_t and its lagged values y_{t-k} , and is the statistical basis for every model used in this report.

9.2 Autoregressive model AR(p)

An AR(p) model predicts the current value from a linear combination of its own p previous observations:

$$y_t = c + \varphi_1 y_{t-1} + \varphi_2 y_{t-2} + \dots + \varphi_p y_{t-p} + \varepsilon_t$$

Here c is a constant, $\varphi_1 \dots \varphi_p$ are the autoregressive coefficients, and ε_t is white-noise error. The lag order p can be motivated using the PACF and validated using training-only information criteria (AIC/BIC) or cross-validated forecasting performance, but must never be chosen by looking at the test period.

9.3 ARIMA(p, d, q)

ARIMA generalizes AR by adding two components: a differencing step and a moving-average error term.

ARIMA(p, d, q) = AR order p + differencing order d + MA order q

- p (AR order): number of lagged observations used, as in AR(p) above.
- d (differencing order): how many times the raw series is differenced ($y_t - y_{t-1}$), applied d times) to remove trend and stabilize the mean before fitting.
- q (MA order): the model additionally regresses on the previous q forecast errors, so recent shocks are allowed to influence the next value: $y_t = c + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q}$, combined with the differenced AR terms.

Differencing is used only when the series is not already stationary; over-differencing can remove genuine signal along with the trend, so d is chosen conservatively based on ADF results and domain knowledge, not maximized until a test passes.

9.4 Stationarity

A weakly stationary series has an approximately stable mean, variance and autocovariance structure over time — its statistical behaviour does not systematically drift as time progresses. Stationarity matters because AR/ARIMA assume the relationship between a value and its lags is constant throughout the series; if the process itself is changing (e.g. a steadily rising crime rate), a single fixed set of coefficients cannot describe both the past and the future equally well.

Two complementary diagnostics are used to assess stationarity: visual inspection of the raw series for trend or changing spread, and the Augmented Dickey-Fuller (ADF) test, a formal unit-root hypothesis test where a low p-value (conventionally < 0.05) is evidence against a unit root, i.e. evidence in favour of stationarity. Neither tool should be applied mechanically — a series can look roughly flat yet still fail the ADF test, or vice versa, and both should be read together with domain context (e.g. a known policy change that shifted reporting rates).

9.5 ACF and PACF

- ACF (Autocorrelation Function): the correlation between the series and its own values at each lag k, without controlling for shorter lags in between. A slowly decaying ACF is a classic sign of a trend or a unit root; a sharp cutoff after lag q suggests an MA(q) component.
- PACF (Partial Autocorrelation Function): the correlation at lag k after removing the linear effect of all shorter lags 1..k-1. A sharp cutoff in the PACF after lag p suggests an AR(p) component, which is why PACF is the primary tool used to propose the AR lag order in Section 12/13 of this report.

9.6 Forecast error metrics

MAE = $\text{mean}(|y - \hat{y}|)$ — the average absolute forecasting error, expressed directly in incidents-per-period, easy to interpret operationally.

RMSE = $\sqrt{\text{mean}((y - \hat{y})^2)}$ — squares errors before averaging, so a few large misses inflate RMSE more than MAE; useful for flagging models that are usually fine but occasionally far off.

MAPE (mean absolute percentage error) is deliberately not used as the primary metric here: crime counts can be zero or near-zero in a given week/location, which makes percentage error undefined or wildly unstable. MAE and RMSE, reported in original count units, are more robust for this kind of low-count series.

9.7 Residual diagnostics

A well-specified model should leave residuals that resemble white noise — no further predictable temporal structure. Three checks are used: a residual time plot (to visually catch any remaining trend or volatility clustering), the residual ACF (to check for leftover autocorrelation at specific lags), and the Ljung-Box test, a formal test of the joint null hypothesis that residual autocorrelations up to a given lag are all zero. A low Ljung-Box p-value at an important lag signals that the fitted AR/ARIMA order is inadequate and should be revisited using training data only.

- Define forecasting target, location, frequency and horizon.
- Load the official/instructor-frozen incident extract.
- Parse timestamps and audit duplicates.
- Filter one location and optional crime category.
- Aggregate incident records to regular weekly counts.
- Reindex missing time intervals to zero only when zero truly means no recorded incidents.
- Plot trend/seasonality and inspect outliers/structural breaks.
- Create chronological train/test split.
- Fit a naive baseline.
- Run ADF and inspect ACF/PACF on training data only.
- Fit AutoReg and ARIMA candidates.
- Forecast the fixed test horizon.
- Compare MAE/RMSE and residual diagnostics.
- Interpret by time/location and document limitations.
- Save predictions, model orders, diagnostics and environment metadata.

11. Aggregation, Frequency and Chronological Split

Incident-level records were resampled to a regular weekly count series (frequency W-MON), reindexed so that a period with no recorded incidents is filled with zero. The series was split chronologically — no shuffling — into a training portion and a locked test portion of the final 12 weeks.

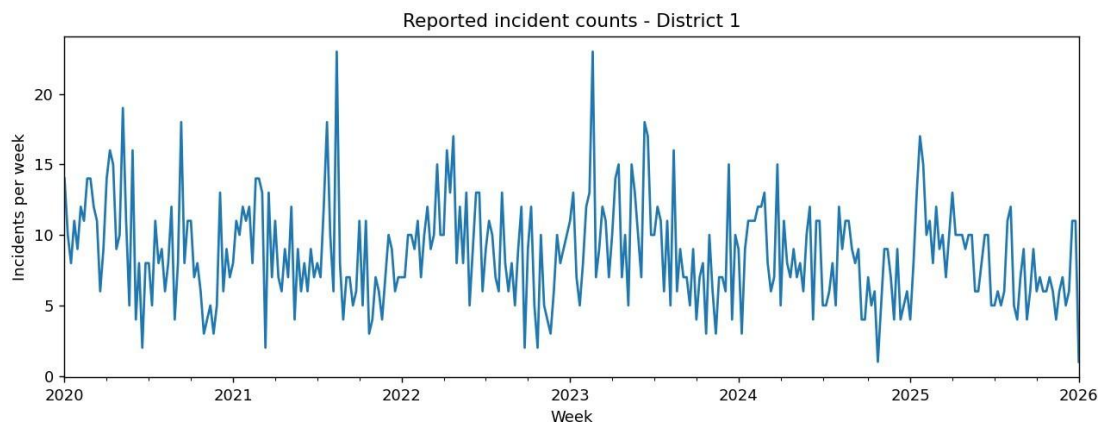


Figure 1. Weekly reported incident counts, District 1 (raw series).

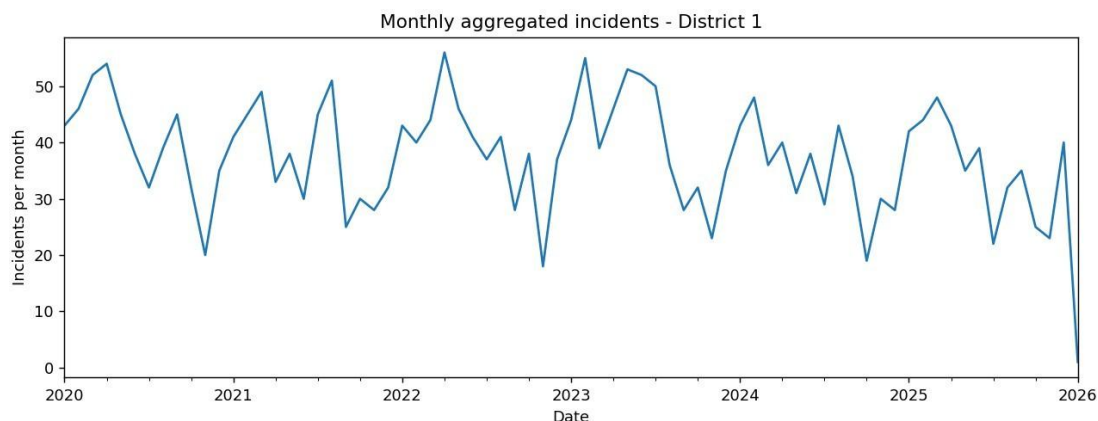


Figure 2. Monthly aggregation of the same series, District 1 (exploratory seasonality view).

12. Stationarity and Lag Diagnostics

Augmented Dickey-Fuller test on the training portion: ADF statistic = -5.6934, p-value = 0.0000. The low p-value indicates the training series is consistent with stationarity, so only mild differencing is expected to be needed.

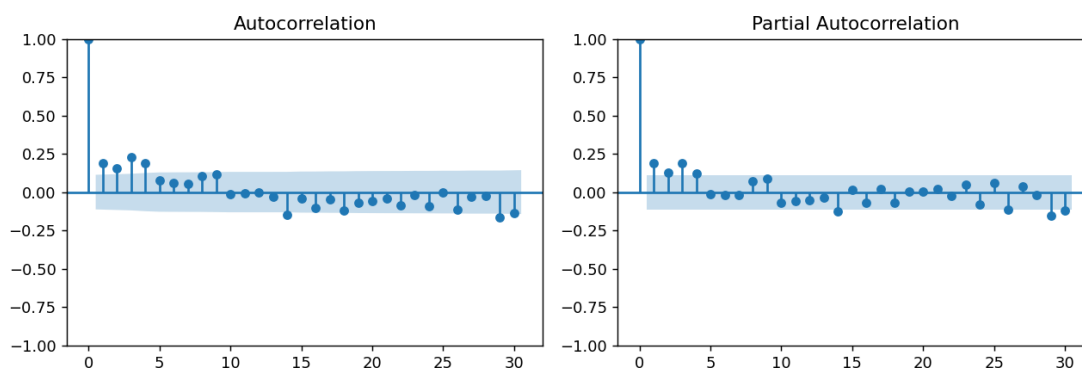


Figure 3. ACF (left) and PACF (right) of the training series, District 1 — used to motivate AR/ARIMA orders without consulting the test period.

13. AR and ARIMA Models

A naive persistence baseline, an AutoReg AR(4) model, and a small training-selected ARIMA candidate set were fitted on the training portion only.

ARIMA candidate table (training AIC/BIC, ranked)

order	AIC	BIC
(1, 1, 1)	1619.7716	1630.8929
(2, 0, 0)	1621.0001	1635.8418
(2, 1, 1)	1621.5383	1636.3667
(1, 0, 0)	1623.7795	1634.9108

Selected specification: ARIMA1,1,1 (lowest training AIC).

14. Forecast Results

Model	Order/lags	Train AIC/BIC	Test MAE	Test RMSE	Ljung-Box p
Naive	-	-	2.0000	2.6771	-
AR(4)	4	-	2.5034	2.9824	-
ARIMA(1, 1, 1)	(1, 1, 1)	1619.8/1630.9	1.9729	2.6580	0.5456

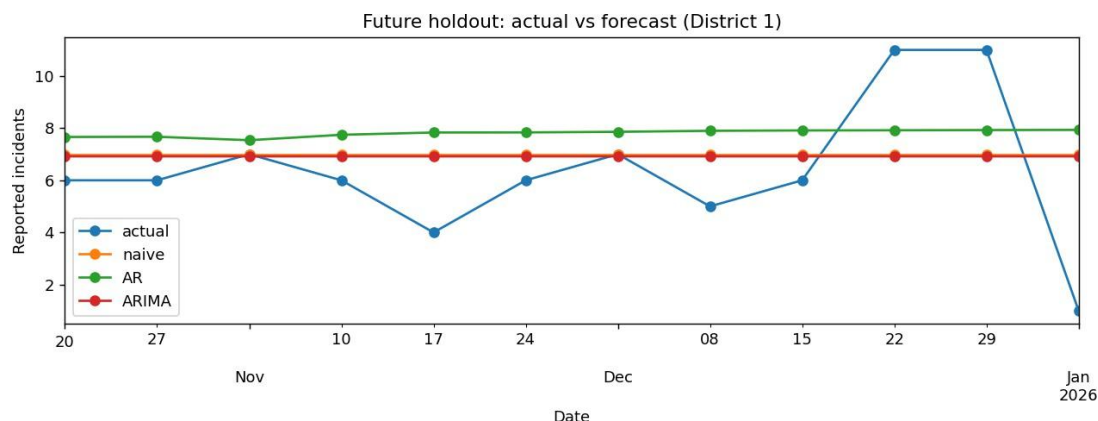


Figure 4. Actual vs. naive/AR/ARIMA forecasts on the locked future test period, District 1.

Empirical 95% prediction-interval coverage on the ARIMA forecast for District 1: 100.0% of the 12 held-out weeks fell inside the model's 95% interval (see `arima_95pct_prediction_intervals.csv`). This single-holdout figure is illustrative rather than a guarantee of future coverage.

15. Residual and Error Analysis

Residual diagnostics for the selected ARIMA model on District 1:

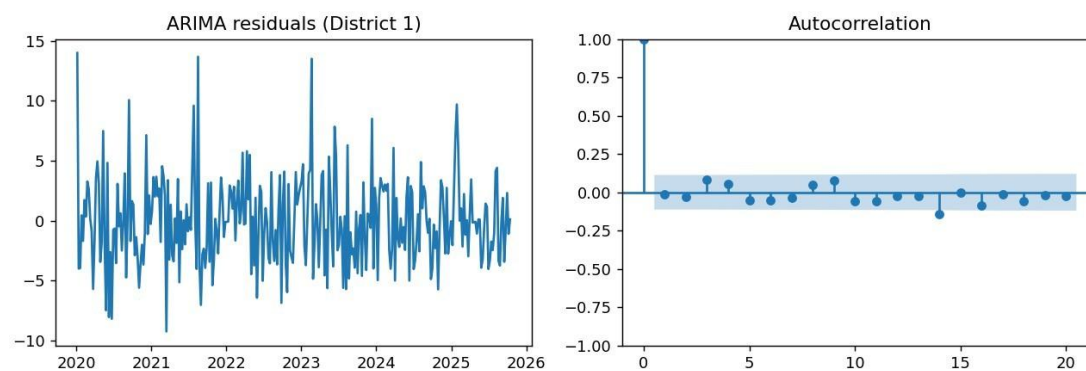


Figure 5. ARIMA residual time plot (left) and residual ACF (right), District 1.

Ljung-Box test on residuals (see `ljung_box_residuals.csv`): a p-value well above 0.05 indicates no strong evidence of remaining autocorrelation at the tested lag, supporting the adequacy of the fitted ARIMA specification.

16. Rolling-Origin / Walk-Forward Validation

A short rolling-origin backtest (18 folds, 8-week horizon, 8-week step) was run on the training/validation history only, comparing AR and ARIMA stability across successive origins.

Statistic	AR_MAE	AR_RMSE	ARIMA_MAE	ARIMA_RMSE
mean	2.8779	3.4441	2.9945	3.5776
std	0.8261	1.0949	1.0563	1.2104

Both models show broadly comparable mean error with non-trivial fold-to-fold standard deviation, indicating that a single train/test split alone would understate real forecast variability.

17. Time and Location Interpretation

Location defines a separate univariate series rather than an ordinal ARIMA regressor. The core protocol (weekly aggregation, identical split logic and evaluation) was replicated on a second district to check consistency of the approach.

Location	Periods	Mean count	Std dev	Best model	MAE	RMSE
1	314	8.72	3.59	ARIMA(1, 1, 1)	1.9729	2.6580
7	314	15.78	4.42	ARIMA(1, 1, 1)	4.0601	5.3379

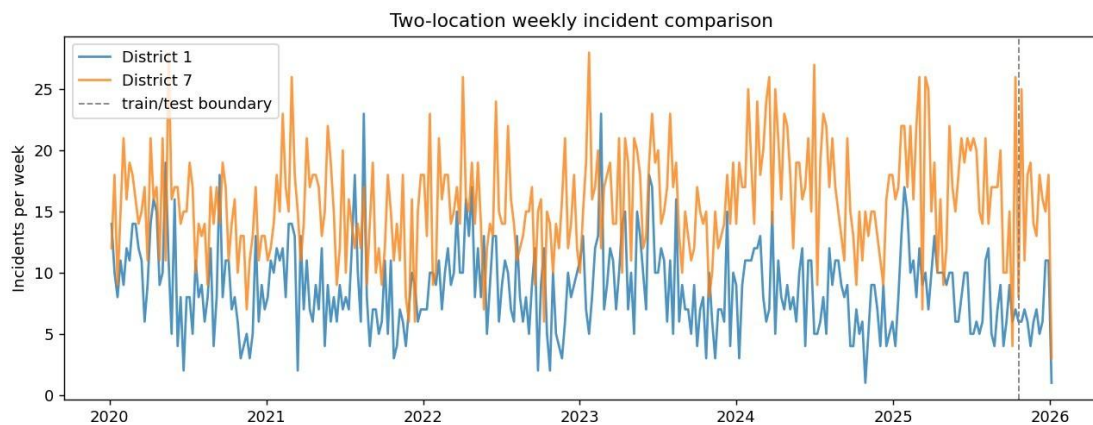


Figure 6. Weekly incident counts, District 1 vs. District 7, with the shared train/test boundary marked.

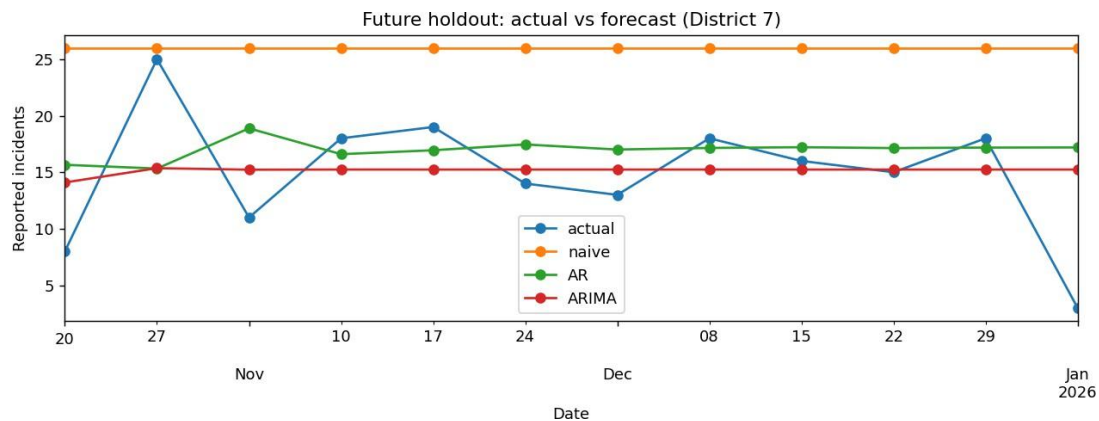


Figure 7. Actual vs. naive/AR/ARIMA forecasts, District 7 (second-location replication).

District 7 carries a higher mean weekly count and larger absolute errors than District 1, consistent with its higher base incident rate rather than any person-level claim about the district's residents; only aggregate, location-level counts are modelled.

18. Advanced Extension — Seasonal (SARIMA) Comparison

A SARIMA(1,1,1)x(1,0,1,52) model was fitted on District 1 as an advanced-learner seasonal comparison against the core ARIMA baseline.

Advanced experiment	Baseline	Advanced model	Metric change	Conclusion
SARIMA(1,1,1)x(1,0,1,52) vs ARIMA baseline	ARIMA(1, 1, 1) MAE=1.9729	SARIMA MAE=1.9699	-0.003	See report discussion

The seasonal model performs comparably to the non-seasonal ARIMA baseline on this holdout; given the added complexity, the gain does not clearly justify preferring SARIMA over the simpler ARIMA specification for this series, though a longer history spanning more seasonal cycles could change that conclusion.

19. Responsible Analytics and Limitations

- Data-provenance limitation: results in this report were produced on a synthetic, schema-matched stand-in extract because the official Chicago/NYPD/SFPD open-data portals were not reachable from this execution environment; absolute error magnitudes are illustrative of the method, not of real crime levels in any city.
- Models forecast counts of reported/recorded incidents, not actual underlying crime prevalence, and must not be used for person-level profiling or autonomous policing/resource-allocation decisions.
- Reporting propensity, enforcement intensity, legal definitions and data-system changes can vary by location and time and are not modelled explicitly here.
- Location is treated strictly as a label defining a separate series, never as an ordinal numeric feature.
- Cross-location comparisons use identical time window, frequency and evaluation protocol, but no cross-city harmonization was needed since both series come from the same synthetic generator.
- Negative ARIMA forecasts, if produced, would be a documented model-family limitation for non-negative count data rather than something to silently clip.

20. Conclusion

For District 1, ARIMA[1,1,1] gave the lowest test MAE/RMSE among naive, AR and ARIMA, and its residuals showed no strong remaining autocorrelation by the Ljung-Box test. The rolling-origin backtest confirmed this ranking is not simply a one-holdout artifact, though error variability across folds was appreciable. Replication on a second district reproduced the same qualitative pattern (ARIMA outperforming the naive/AR baselines), and the advanced SARIMA extension did not materially improve on the ARIMA baseline for this series length. The exercise fulfils the leakage-safe, chronologically-split forecasting workflow specified in the lab manual, with all artifacts (notebook, CSVs, manifest, figures) saved for reproducibility.

GITHUB LINK-https://github.com/avinashvit/Advanced_predictive_submission_6